

SYLLABUS :-

Rock Mechanics Laboratory x Preparation of rock samples for uniaxial compressive strength UCS, tensile strength and punch shear test as per international society of rock mechanics ISRM standards. x Determination of rock quality designation RQD and core recovery factor for borehole logs. x Determination of bulk density, specific gravity, porosity, void ratio and hydraulic conductivity of a rock sample. x Determination of UCS, Young's Modulus, Poisson's ratio of rock sample by universal testing machine; UCS by point load strength index test; Cohesion and angle of internal friction of intact rock sample by i oblique shear test; and ii Tri-axial test; Cohesion and Angle of internal friction by direct shear test; Shear strength by punch shear test, and weathering characteristics of rock sample. x Determination of abrasivity by micro-bit drilling and Cerchar abrasivity test. x Non-Destructive test on rock sample for UCS, Young's modulus of elasticity and Poisson's ratio.

0-0-3 Reference Books/ Web-link: 1. Rock Mechanics Laboratory Manual, Department of Mining Engineering, IIT Kharagpur. 2. R. Ulusay eds. 2015 The ISRM Suggested Methods for Rock Characterization, Testing and Monitoring: 2007-2014, Springer, ISBN : 978-3-319-07712-3. 3. R. Ulusay, J. A. Hudson eds. 2009: The Complete ISRM Suggested Methods for Rock Characterisation, Testing and Monitoring: 1974 - 2006, Compiled by ISRM Turkish National Group. www.isrm.net